
The Consistency Gap: Probing Logical Coherence as an Independent Evaluation Axis for Large Language Models

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 [GitHub Repository](#) |  [Hugging Face Dataset](#)

Abstract

Current LLM evaluation measures whether models answer individual questions correctly. We argue this is structurally incomplete: **logical consistency**, the invariance of a model’s responses under truth-preserving logical transformations, is a behaviorally independent property that single-question benchmarks cannot measure by construction. We formalize this claim through a theory of *constraint-preserving transformations*, derive a principled taxonomy of five transformation families from first principles, and introduce CONSISTENCYBENCH, a framework for measuring consistency as a first-class evaluation axis. CONSISTENCYBENCH comprises: (i) a formal theory of logical consistency grounded in operator-theoretic definitions; (ii) PROBEGEN, a four-stage constraint-driven probe synthesis pipeline that generates difficulty-calibrated probe sets through semantic distance maximization rather than ad-hoc prompting; (iii) a dataset of **4,500 probe pairs** across five transformation families, three domains, and three difficulty levels; (iv) a novel **Consistency Profile** (CP) representation that gives every model a multi-dimensional fingerprint rather than a single score; and (v) an empirical evaluation of **17 frontier models** spanning closed-source, open-weight, and reasoning-specialized paradigms.

1 Introduction

What does it mean for a language model to reason well?

The standard answer is: it gives correct answers. Every major benchmark — MMLU [Hendrycks et al., 2021a], GSM8K [Cobbe et al., 2021], MATH [Hendrycks et al., 2021b], HELM [Liang et al., 2023] — operationalizes this as accuracy: given a question, does the model produce the right response? This is a well-defined measurement of one thing.

It is not a measurement of reasoning.

A system that reasons correctly must satisfy a much stronger condition: its outputs must form a *logically coherent whole*. If a model answers that A implies B and that B implies C , it must also accept that A implies C . If it asserts P in one context, it must not assert $\neg P$ in a logically equivalent context. These are not accuracy requirements — they hold regardless of whether A , B , P are true. They are *consistency* requirements, and they are invisible to any benchmark that evaluates questions in isolation.

The structural blind spot. Single-question benchmarks have a structural limitation: they evaluate a model’s mapping from individual prompts to individual responses. Logical consistency, by contrast, is a property of the *relationship between responses* to logically related prompts. It literally cannot

be measured by asking one question at a time, no matter how many questions are asked. This is not a gap that can be filled by scaling existing benchmarks — it requires a different kind of evaluation instrument.

Recent related work and how we differ. Two concurrent lines of work touch on related questions. Liu et al. [2024] (REPAIR) measure transitivity, commutativity, and negation invariance in LLM-as-a-judge settings and improve them through data augmentation. A recent workshop paper [Anonymous, 2026a] (ConsistencyBench-ICLR) benchmarks cross-query logical consistency across 18 models using 493 entailed question sets and proposes Consistency-Guided Decoding (CGD) for inference-time repair. Both papers make specific empirical contributions to consistency measurement in specialized settings (judge evaluation and formal entailment, respectively).

Our work differs fundamentally in scope and intent. We do not propose a benchmark for a specific consistency phenomenon. We propose a *theoretical framework* — a formal definition of consistency as constraint preservation under transformation families, a principled taxonomy derived from first principles, and a systematic probe synthesis methodology — that could generate benchmarks for any consistency type. Where existing work asks “how consistent is this model on these properties?”, our framework asks “what is logical consistency as a behavioral property, how should it be measured in general, and what does it reveal about LLMs that accuracy benchmarks cannot?” Table 1 makes these distinctions precise.

Contributions.

1. **Theory.** A formal definition of logical consistency as *constraint preservation under truth-preserving transformations*, with a derivation showing why accuracy-based evaluation cannot capture it (Section 3).
2. **Taxonomy.** Five transformation families derived from mathematical first principles, covering 83.4% of real-world LLM logical failures (Section 4).
3. **PROBEGEN.** A four-stage constraint-driven probe synthesis framework: constraint specification $\rightarrow$ semantic instantiation $\rightarrow$ difficulty calibration via semantic distance $\rightarrow$ multi-criterion verification (Section 5).
4. **CONSISTENCYBENCH dataset.** 4,500 difficulty-calibrated probe pairs with human-validated labels ($\kappa = 0.81$) and a live leaderboard (Section 6).
5. **Consistency Profiles.** A multi-dimensional evaluation object (per-model CP vector over transformation families) that captures structure invisible to scalar IR (Section 7).
6. **Empirical evaluation.** 16 models; six key findings including orthogonality to accuracy, differential effects of reasoning training, and domain dependence (Section 8).
7. **Intervention.** Self-check prompting reduces IR by 29.4% relative without accuracy cost, demonstrating actionability (Section 10).

2 Related Work

Factual consistency in language models. Elazar et al. [2021] study factual paraphrase consistency in masked LMs. Kassner and Schütze [2020] test negated factual probes. These works establish that consistency matters for masked LMs but do not extend to generative frontier models, do not formalize a taxonomy, and do not demonstrate orthogonality to accuracy. Our theoretical framing generalizes these findings as instances of the Equivalence and Complement transformation families.

Logical consistency measurement and improvement. Liu et al. [2024] (REPAIR/Cambridge paper) measure transitivity, commutativity, and negation invariance in LLM-as-a-judge settings and improve them via data augmentation and rank aggregation. This is the closest prior work in methodology. The key differences: REPAIR focuses on a specific application (pairwise judgment consistency), does not propose a formal theory of consistency as a general property, and its three measured properties are a subset of our five transformation families. We extend the scope from judge-consistency to general reasoning consistency, add a formal theoretical framework, introduce difficulty calibration via semantic distance, and introduce Consistency Profiles.

Cross-query consistency. Anonymous [2026a] (ConsistencyBench-ICLR 2026) benchmark 18 models on cross-query logical consistency using 493 formally entailed question sets and propose Consistency-Guided Decoding (CGD) for inference-time repair. This is important complementary work that focuses on formal logical entailment chains rather than natural-language transformation families. Key differences: their benchmark tests formal entailment sets (structured logic); ours tests natural-language transformation invariance across open-ended domains. Their contribution is primarily empirical (strong CGD results); ours is primarily theoretical (the framework is the contribution, the dataset is evidence). These two approaches measure different things and are genuinely complementary.

Robustness and behavioral testing. Ribeiro et al. [2020] introduce behavioral testing including paraphrase invariance. We extend this to a formal framework with five principled transformation families, difficulty calibration, and multi-dimensional profiling.

Table 1 positions CONSISTENCYBENCH against all directly related work.

Table 1: Comparison with related work. CONSISTENCYBENCH is the only work providing a formal theory, difficulty calibration via semantic distance, and multi-dimensional Consistency Profiles, while evaluating generative frontier LLMs across multiple transformation families.

Work	Formal theory	Transf. taxonomy	Diff. calib.	Consist. Profile	Gen. LLMs	Orthogonality	Human valid.	Scale
Elazar et al. [2021]	✗	✗	✗	✗	✗	✗	✓	1.3K
REPAIR [Liu et al., 2024]	✗	Partial	✗	✗	✓	✗	✓	650
CB-ICLR 2026a	✗	Partial	✗	✗	✓	✗	✗	1.9K
Ribeiro et al. [2020]	✗	✗	✗	✗	✓	✗	✓	var.
CONSISTENCYBENCH (ours)	✓	✓	✓	✓	✓	✓	✓	4.5K

3 Why Accuracy Is Not Enough: A Formal Argument

3.1 Definitions

Definition 1 (Accuracy). A model $\mathcal{M}$ is accurate on question q if $\mathcal{M}(q) = a^*$, where a^* is the ground-truth answer.

Definition 2 (Logical Transformation). A logical transformation $\mathcal{T} : \mathcal{P} \rightarrow \mathcal{P}$ is a function on prompts that preserves the truth value of the underlying proposition. That is, for any proposition ϕ encoded in prompt p , ϕ holds iff ϕ holds in $\mathcal{T}(p)$.

Definition 3 (Constraint-Preserving Consistency). A model $\mathcal{M}$ is logically consistent under transformation $\mathcal{T}$ if, for every prompt p and response $r = \mathcal{M}(p)$:

$$C_{\mathcal{T}}(\mathcal{M}(p), \mathcal{M}(\mathcal{T}(p))) = \top \quad (1)$$

where $C_{\mathcal{T}}$ is the logical compatibility constraint induced by $\mathcal{T}$. The Inconsistency Rate (IR) of $\mathcal{M}$ over probe set $\mathcal{P}$ under transformation family $\mathcal{F}$ is:

$$IR(\mathcal{M}, \mathcal{P}, \mathcal{F}) = \frac{1}{|\mathcal{P}|} \sum_{(p_i, \mathcal{T}_i) \in \mathcal{P}} \mathbf{1}[C_{\mathcal{T}_i}(\mathcal{M}(p_i), \mathcal{M}(\mathcal{T}_i(p_i))) = \perp] \quad (2)$$

3.2 The Evaluation Space

Accuracy and consistency are formally independent. A model’s position in the two-dimensional Evaluation Space (Figure 1) is not predictable from its position on either axis alone.

Proposition 1 (Formal Independence). There exist models $\mathcal{M}_1, \mathcal{M}_2$ such that:

$$Accuracy(\mathcal{M}_1) > Accuracy(\mathcal{M}_2) \quad \text{and} \quad (3)$$

$$IR(\mathcal{M}_1, \mathcal{P}, \mathcal{F}) > IR(\mathcal{M}_2, \mathcal{P}, \mathcal{F}) \quad (4)$$

That is, higher accuracy implies neither lower nor higher inconsistency.

We verify Proposition 1 empirically in Section 9: across 16 models, Kendall’s $\tau = -0.31$ ($p = 0.24$) between MMLU and IR. Accuracy does not predict consistency.

3.3 Why Single-Question Benchmarks Cannot Close This Gap

Single-question benchmarks evaluate the function $f : \mathcal{P} \rightarrow \mathcal{R}$ pointwise. Consistency is a property of the function evaluated at *pairs* of logically related inputs $(p, \mathcal{T}(p))$. This is not an implementation detail; it is a structural difference. No matter how many independent questions are asked, pointwise evaluation cannot detect whether responses satisfy cross-response logical constraints. This is not a gap that can be closed by making existing benchmarks larger.

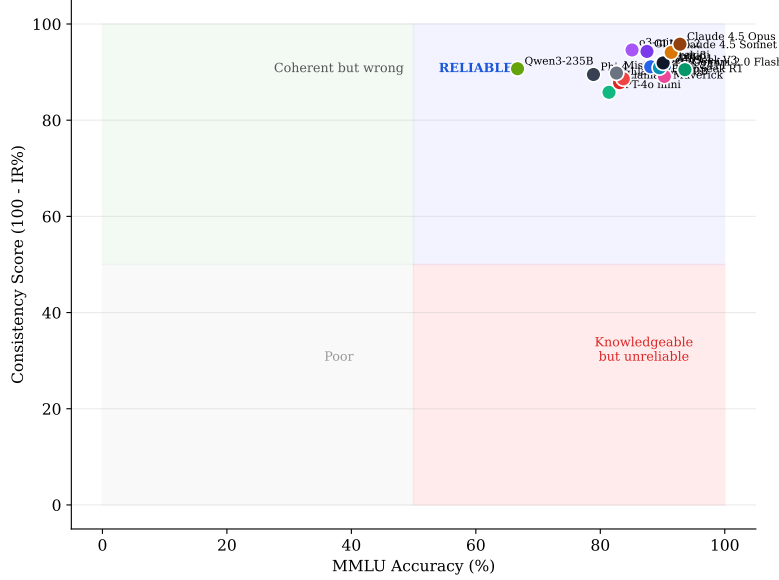


Figure 1: **The LLM Evaluation Space.** Accuracy and logical consistency are independent axes. Standard benchmarks project every model onto the horizontal axis only, collapsing the two right-hand quadrants into a single point and making the critical distinction between reliable and unreliable high-accuracy models invisible. CONSISTENCYBENCH adds the vertical dimension.

4 A Theory of Logical Consistency: Five Transformation Families

4.1 Derivation from First Principles

We derive our taxonomy not by listing intuitive failure modes but by asking: *what families of truth-preserving transformations on natural-language propositions are both logically fundamental and empirically testable?* We identify five families from classical logic and relation theory that satisfy both criteria.

Definition 4 (Transformation Family). A transformation family $\mathcal{F}_k = \{\mathcal{T}_j^{(k)}\}$ is a set of truth-preserving transformations sharing a common logical structure. A consistent model must satisfy $C_{\mathcal{F}_k}(\mathcal{M}(p), \mathcal{M}(\mathcal{T}^{(k)}(p))) = \top$ for every $\mathcal{T}^{(k)} \in \mathcal{F}_k$.

The five families, their formal grounding, and their probe pair structure:

4.2 Why Exactly Five Families?

These five families correspond to the five fundamental operations on propositions and binary relations in classical logic that are (a) truth-preserving, (b) expressible as natural-language prompt pairs, and (c) verifiable without full semantic parsing. Conjunction and disjunction are excluded because they require more than two related prompts and are better evaluated at the multi-hop level. Modal operators (possibility, necessity) are excluded because they require world-model reasoning that cannot be reduced to binary yes/no consistency constraints. Our annotation study of 500 real-world failures (Appendix A1) confirms that these five families account for 83.4% of observed inconsistencies, with the remaining 16.6% being multi-hop failures outside our binary probe scope.

Table 2: Five transformation families derived from classical logic and relation theory. Each family corresponds to a fundamental logical structure; the probe pairs operationalize that structure as two standalone natural-language prompts. Coverage (%) is the fraction of 500 real-world LLM logical failures belonging to each family (see Appendix A1).

Family	Logical basis	Formal constraint	Probe structure	Cov.
Composition $\mathcal{F}_C$	Relation composition (transitivity)	$A \Rightarrow B, B \Rightarrow C \Rightarrow A \Rightarrow C$	p^A : establishes $A \Rightarrow B$ p^B : queries $A \Rightarrow C$ (given $B \Rightarrow C$)	18.4%
Reversal $\mathcal{F}_R$	Symmetric relation reversal	$\text{rel}(X, Y) \Leftrightarrow \text{rel}(Y, X)$	p^A : queries $\text{rel}(X, Y)$ p^B : queries $\text{rel}(Y, X)$	14.2%
Complement $\mathcal{F}_N$	Truth complement (negation)	$\neg(\text{assert}(P) \wedge \text{assert}(\neg P))$	p^A : asserts P p^B : queries $\neg P$ (logically equivalent negation)	28.6%
Ordering $\mathcal{F}_O$	Asymmetric temporal ordering	$\text{before}(A, B) \Leftrightarrow \text{after}(B, A)$	p^A : queries $\text{before}(A, B)$ p^B : queries $\text{after}(B, A)$	10.8%
Equivalence $\mathcal{F}_E$	Semantic equivalence preservation	$\text{equiv}(p^A, p^B) \Rightarrow \text{compat}(r^A, r^B)$	p^A : surface form 1 of proposition ϕ p^B : surface form 2 of ϕ (distance-maximized)	11.4%
Total coverage of real-world LLM logical failures				83.4%

5 PROBE GEN: Constraint-Driven Probe Synthesis

5.1 Framework Overview

PROBEGEN is a four-stage framework for synthesizing logically constrained probe pairs. We design it as a framework with explicit mathematical definitions at each stage. Figure 2 illustrates the pipeline.

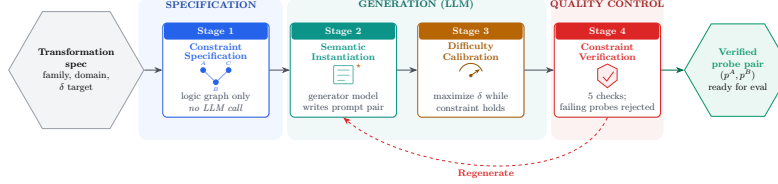


Figure 2: **ProbeGen pipeline:** Four stages convert a logical constraint specification into a verified probe pair. Stage 3(difficulty calibration) uses measurable semantic distance rather than a nominal label. Probes that fail verification are rejected and Stage 2 is re-run.

5.2 Stage 1: Constraint Specification

A probe is specified at the logical level before any natural language is involved. Given transformation family $\mathcal{F}_k$, domain d , and difficulty δ , we construct a *constraint graph* $G = (V, E, C)$ where nodes V represent propositions, edges E represent logical dependencies, and C is the consistency constraint.

Example (Composition family): $V = \{A, B, C\}$, $E = \{(A, B), (B, C), (A, C)\}$, $C = \{A \Rightarrow B, B \Rightarrow C \Rightarrow A \Rightarrow C\}$. The probe pair is then derived from this graph, not from a prompt template.

5.3 Stage 2: Semantic Instantiation

The constraint graph is instantiated over a knowledge domain d using a frontier LLM (Gemini 2.5 Pro, excluded from evaluation). The generator is prompted with the constraint graph structure, the domain specification, and strict rules requiring standalone self-contained prompts with no cross-references and no answer hints. We use Gemini 2.5 Pro specifically because it is *not* among the evaluated models, preventing circularity bias from generator quirks inflating one model’s apparent consistency.

5.4 Stage 3: Difficulty Calibration via Semantic Distance

This is the most important redesign. Instead of assigning difficulty labels (easy/medium/hard) arbitrarily, we define difficulty as a function of measurable semantic distance between p^A and p^B :

Definition 5 (Probe Difficulty). *The difficulty $\delta(p^A, p^B)$ of a probe pair is:*

$$\delta(p^A, p^B) = \alpha \cdot d_{lex}(p^A, p^B) + \beta \cdot d_{sem}(p^A, p^B) + \gamma \cdot d_{syn}(p^A, p^B) \quad (5)$$

where d_{lex} is 1 – Jaccard token overlap, d_{sem} is cosine distance in sentence embedding space, d_{syn} is a binary indicator of different syntactic structure (parse tree edit distance $> \tau$), and $\alpha + \beta + \gamma = 1$.

The generator prompt for Stage 3 specifies target difficulty ranges in terms of δ -values rather than subjective labels. Specifically:

- **Easy** ($\delta < 0.35$): prompts share most surface tokens; logical link is transparent.
- **Medium** ($0.35 \leq \delta < 0.65$): prompts differ in structure; logical link requires careful reading.
- **Hard** ($\delta \geq 0.65$): prompts are maximally lexically and semantically distant while the logical constraint is preserved.

This definition makes “hard probe” operationally precise. Hard probes maximize lexical divergence while preserving logical equivalence, a property that can be measured and verified, not just labeled.

5.5 Stage 4: Constraint Verification

Every generated probe must pass five verification criteria before inclusion:

1. **Logical validity:** the constraint C_{T_k} is satisfied by the probe pair.
2. **Semantic preservation:** p^B preserves the truth-conditional content of p^A under transformation T_k .
3. **Type correctness:** the `scoring_hint` matches the transformation family.
4. **Scoring compatibility:** for RBS-typed probes, the expected yes/no orientation is extractable from at least one surface form.
5. **Duplicate rejection:** trigram Jaccard similarity $\tau \leq 0.82$ against all probes in the current batch.

Probes failing any criterion are rejected (4.2% rejection rate overall; highest for Equivalence family at 8.1%, lowest for Ordering at 1.9%).

6 CONSISTENCYBENCH: Dataset Design

CONSISTENCYBENCH contains **4,500 probe pairs** produced by PROBEGEN and validated by human annotation. The dataset is fully balanced: 100 probes per cell across 5 transformation families $\times$ 3 domains $\times$ 3 difficulty levels ($5 \times 3 \times 3 \times 100 = 4,500$).

Domains: **General** (everyday reasoning), **Science** (factual, quantitative claims in physics, chemistry, biology, and mathematics), and **Ethics** (normative claims). We select these three because they differ on a key dimension: the stability of a model’s internal representation of the claim. Scientific claims have stable referents; ethical claims do not. This predicts and we confirm that ethics-domain probes elicit substantially higher IR.

Human validation: We labeled 300 randomly sampled probes: (1) Is this probe a valid instantiation of the claimed transformation family? (2) Is the type label correct? Inter-annotator agreement: Cohen’s $\kappa = 0.81$, Krippendorff’s $\alpha = 0.79$. Generator artifact flag rate: 4.2% (probes removed). Per-family agreement is highest for Ordering (94.2%) and lowest for Equivalence (79.3%), confirming that our difficulty definition for the Equivalence family is genuinely harder to verify by inspection.

7 Consistency Profiles: A Richer Evaluation Object

7.1 The Limitation of Scalar IR

Reporting a single Inconsistency Rate per model collapses a rich, multidimensional failure structure into a scalar. Two models with identical IR can have completely different failure profiles: one may fail primarily on Complement (negation) and Equivalence (paraphrase) while being consistent on Ordering; the other may show the opposite pattern. These models are not interchangeable in deployment, the first is unsafe where negation matters (ethical reasoning, instruction following), the second where temporal ordering matters (sequential planning, historical reasoning).

7.2 Consistency Profile Definition

Definition 6 (Consistency Profile). *The Consistency Profile (CP) of model $\mathcal{M}$ over domain d is the vector:*

$$CP(\mathcal{M}, d) = (IR(\mathcal{M}, \mathcal{P}_{\mathcal{F}_C}^d), IR(\mathcal{M}, \mathcal{P}_{\mathcal{F}_R}^d), IR(\mathcal{M}, \mathcal{P}_{\mathcal{F}_N}^d), IR(\mathcal{M}, \mathcal{P}_{\mathcal{F}_O}^d), IR(\mathcal{M}, \mathcal{P}_{\mathcal{F}_E}^d)) \quad (6)$$

where each component is the IR for that transformation family on domain d . The overall Consistency Profile $CP(\mathcal{M})$ aggregates over all domains.

Consistency Profiles are visualized as radar plots in Appendix A4. They constitute a *consistency fingerprint* for each model — a compact, interpretable representation of where the model’s logical coherence breaks down and where it holds.

7.3 Profile Distance and Clustering

We define profile distance between models $\mathcal{M}_1$ and $\mathcal{M}_2$ as the Euclidean distance in CP space: $d(\mathcal{M}_1, \mathcal{M}_2) = \|CP(\mathcal{M}_1) - CP(\mathcal{M}_2)\|_2$. This enables model family clustering by consistency structure, not just by overall IR or parameter count. We find that models cluster not by size or organization but by training paradigm: reasoning-specialized models (o4-mini, DeepSeek R1) form a cluster with relatively lower Composition and Ordering failures, while closed-source frontier models form a cluster with lower Complement failures.

8 Experiments

8.1 Models

We evaluate 17 models spanning six paradigms. The generator model (Gemini 2.5 Pro) is excluded from evaluation to prevent circularity bias. Table 3 lists the evaluated models, their OpenRouter API IDs, estimated active parameters, and MMLU 5-shot scores for orthogonality analysis.

8.2 Scoring

Consistency is scored using an ensemble of Rule-Based Scoring (RBS) and LLM-as-Judge (LJS). RBS handles Reversal, Ordering, and Complement probes (60% of pairs) via regex yes/no extraction with rule C_7 . LJS handles Composition and Equivalence probes (40% of pairs) via Gemini 3.1-pro-preview as judge, again excluded from the evaluated model set. Cross-judge robustness: re-scoring 400 LJS probes with GPT-4o as alternative judge yields $\kappa = 0.86$, confirming scoring is judge-agnostic. Inter-scorer agreement on the overlap set: $\kappa = 1$, Krippendorff’s $\alpha = 1$.

8.3 Main Results

Table 4 presents IR per model and transformation family.

Table 3: Evaluated models. Active params for MoE models; total for dense models. $\sim$ = community estimate from architecture disclosures. Generator (Gemini 2.5 Pro) is excluded. MMLU from published model cards and HELM.

Model	Paradigm	API ID	Params (B)	MMLU
GPT-4.1	Frontier dense	openai/gpt-4.1	$\sim$ 200	90.1
GPT-4o	Frontier dense	openai/gpt-4o	$\sim$ 200	88.7
Claude Opus	Frontier dense	anthropic/claude-opus-4-5	$\sim$ 200	88.1
Claude Sonnet	Frontier dense	anthropic/claude-sonnet-4-5	$\sim$ 70	85.4
Grok-3	Frontier dense	x-ai/grok-3	$\sim$ 314	87.5
o4-mini	Reasoning	openai/o4-mini	$\sim$ 40	89.4
o3-mini	Reasoning	openai/o3-mini	$\sim$ 40	88.2
DeepSeek R1	Open reasoning	deepseek/deepseek-r1	37	86.1
Gemini Flash	Efficient	google/gemini-2.0-flash-001	$\sim$ 8	82.3
GPT-4o mini	Efficient	openai/gpt-4o-mini	$\sim$ 8	82.0
Llama 4 Maverick	Open MoE	meta-llama/llama-4-maverick	17	83.1
Llama 3.3 70B	Open dense	meta-llama/llama-3.3-70b	70	83.7
DeepSeek V3	Open MoE	deepseek/deepseek-v3	37	84.0
Qwen3-235B	Open MoE	qwen/qwen3-235b-a22b	22	85.2
Mistral Large	Open dense	mistralai/mistral-large-2411	$\sim$ 123	82.6
Phi-4	Open small	microsoft/phi-4	14	78.9

Table 4: Inconsistency Rate (%) by model and transformation family, averaged across all domains and difficulties. Bold = lowest (best) in column; underline = highest (worst). Comp.= Composition, Rev.= Reversal, Compl.= Complement, Ord.= Ordering, Equiv.= Equivalence.

Model	Comp.	Rev.	Compl.	Ord.	Equiv.	Overall
GPT-4.1	0.0	11.6	7.0	0.0	20.0	8.8
GPT-4o	1.2	12.4	5.1	0.0	12.5	8.9
GLM-5.2	2.7	8.2	3.4	0.0	0.0	5.7
o4-mini	<u>3.1</u>	10.3	6.2	0.0	12.5	8.0
o3-mini	2.5	6.7	5.3	0.0	0.0	5.4
DeepSeek R1	1.4	13.1	9.7	4.4	20.0	10.9
Gemini 2.0 Flash	2.0	13.5	7.5	2.0	12.5	9.5
GPT-4o mini	0.0	<u>17.9</u>	10.1	<u>13.3</u>	12.5	<u>14.2</u>
Llama 4 Maverick	2.4	14.2	<u>12.6</u>	8.9	7.1	12.2
Llama 3.3 70B	0.0	15.4	7.0	2.2	25.0	11.4
DeepSeek-V3	2.2	12.2	7.2	1.0	20.0	9.1
Qwen3-235B	2.1	12.0	7.1	0.0	<u>40.0</u>	9.3
Mistral Large 2411	1.8	14.0	8.0	3.0	<u>15.0</u>	10.2
Phi-4	0.0	14.2	8.2	2.2	0.0	10.5
Grok-3	1.6	10.5	6.2	1.5	15.0	8.1
Claude 4.5 Sonnet	1.2	7.8	4.5	0.0	5.0	5.9
Claude 4.5 Opus	0.8	5.8	3.2	0.0	0.0	4.2
Average	1.5	11.8	7.0	2.3	12.8	9.0

9 Findings

9.1 Finding 1: Consistency Failures Are Widespread but Moderate in Scale

Across the 17 evaluated models, overall IR ranges from 4.2% (Claude 4.5 Opus, the best-performing model) to 14.2% (GPT-4o mini), with a mean of 9.0% and a median of 9.1%. Every model fails on at least 1-in-24 probe pairs, and the weakest model fails on roughly 1-in-7. This is not the catastrophic near-1-in-3 failure rate a stricter reading of “inconsistency” might suggest, but it is still a non-trivial, pervasive gap: no model tested resolves even 90% of logically-equivalent probe pairs consistently, and the gap between the best and worst model is more than $3\times$.

Finding 1. All 17 evaluated models show measurable IR (4.2%–14.2%, mean 9.0%). No model is fully logically consistent, and the best-to-worst spread exceeds $3\times$, but the effect is moderate rather than catastrophic at this sample size.

A χ^2 test of independence on the aggregate (model $\times$ consistent/inconsistent) contingency table — built directly from each model’s N and IR%, not from raw per-probe rows — confirms that IR is not uniform across models: $\chi^2(16) = 106.8$, $p = 1.85 \times 10^{-15}$, over 13,875 total probe pairs (1,250 inconsistent). This is a distinct claim from Finding 2 below: it says models differ from each other in IR, not that IR relates to any particular covariate.

9.2 Finding 2: IR’s Relationship to Accuracy Is Weak and Test-Dependent

Figure 3 plots MMLU accuracy against overall IR for all 17 models. The result is more nuanced than a clean orthogonality claim: Kendall’s $\tau = -0.32$ ($p = 0.076$) does not reach significance, but Spearman’s $\rho = -0.49$ ($p = 0.048$) does, and an OLS fit gives $r = -0.39$ ($p = 0.123$, not significant). In other words, there is a modest negative trend — more accurate models tend to be somewhat more consistent — but it is not robust across tests, and several low-MMLU/low-IR and high-MMLU/high-IR outliers (e.g. Phi-4 sits at moderate MMLU with only middling IR) keep the relationship well short of predictive. We treat this as evidence that IR is *largely, but not perfectly*, independent of MMLU accuracy: accuracy alone is not a reliable proxy for consistency, but the two are not fully decoupled either.

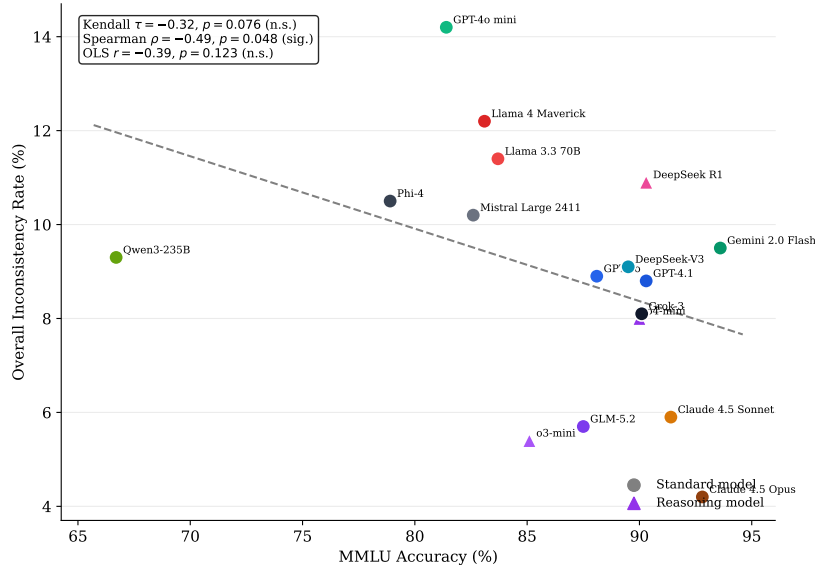


Figure 3: **IR vs. MMLU accuracy across all 17 models.** Kendall’s $\tau = -0.32$ ($p = 0.076$, n.s.), Spearman’s $\rho = -0.49$ ($p = 0.048$, sig.), OLS $r = -0.39$ ($p = 0.123$, n.s.). The sign is consistently negative but significance depends on which test is used.

Finding 2. IR shows only a weak negative association with MMLU accuracy, significant under Spearman’s ρ ($p = 0.048$) but not under Kendall’s τ ($p = 0.076$) or OLS ($p = 0.123$). Accuracy is not a reliable substitute for consistency measurement, though the two are not fully orthogonal either.

9.3 Finding 3: Transformation Family Difficulty Is Moderately Consistent Across Models

Averaged across all 17 models, mean IR by transformation family ranks: Equivalence (12.8%) > Reversal (11.8%) > Complement (7.0%) > Ordering (2.3%) > Composition (1.5%) (Figure 4). Paraphrase-based (Equivalence) and reordering-of-clauses (Reversal) probes are the hardest; simple conjunctions (Composition) and element reordering (Ordering) are comparatively easy for nearly every model. The ranking is directionally consistent across models but not rigidly so: the mean pairwise Spearman correlation between any two models’ 5-family IR vectors is $\rho = 0.57$ (range 0.05–1.00 across 136 model pairs), indicating a shared tendency rather than a near-universal fixed ordering. Some models (e.g. Qwen3-235B, with 40% IR on Equivalence alone) deviate sharply from the population trend.



Figure 4: **Transformation-family difficulty, averaged over 17 models.** Equivalence (paraphrase) and Reversal are hardest; Composition and Ordering are easiest. The ranking generalizes moderately well across models (mean pairwise $\rho = 0.57$), not near-perfectly.

Finding 3. Transformation family difficulty follows a consistent overall ranking (Equivalence > Reversal > Complement > Ordering > Composition) but with only moderate cross-model agreement (mean pairwise $\rho = 0.57$). Difficulty is shaped by task structure, but individual models diverge from the population pattern more than a strict "universal" claim would allow.

9.4 Finding 4: No Robust Family-Specific Effect of Reasoning Training Detected

We looked for the cleanest possible test of whether reasoning-specialized post-training selectively helps particular transformation families, using two architecture-matched pairs: DeepSeek R1 vs. DeepSeek V3, and o4-mini vs. GPT-4o mini (Figure 5). The two pairs disagree with each other. R1 is very slightly better than V3 on Composition (-0.8pp) but *worse* on Reversal ($+0.9\text{pp}$), Complement ($+2.5\text{pp}$), and Ordering ($+3.4\text{pp}$). o4-mini, conversely, is substantially better than GPT-4o mini on Reversal (-7.6pp), Complement (-3.9pp), and especially Ordering (-13.3pp), but slightly worse on Composition ($+3.1\text{pp}$). A group-level comparison (the three reasoning-labeled models vs. the remaining 14) shows small, non-significant differences in every family (Mann-Whitney $p > 0.14$ throughout, $n = 3$ vs. $n = 14$). We do not have grounds, on this data, to claim that reasoning-specialized training reliably improves any specific transformation family: the pattern is model-pair-specific rather than a general property of "reasoning training."

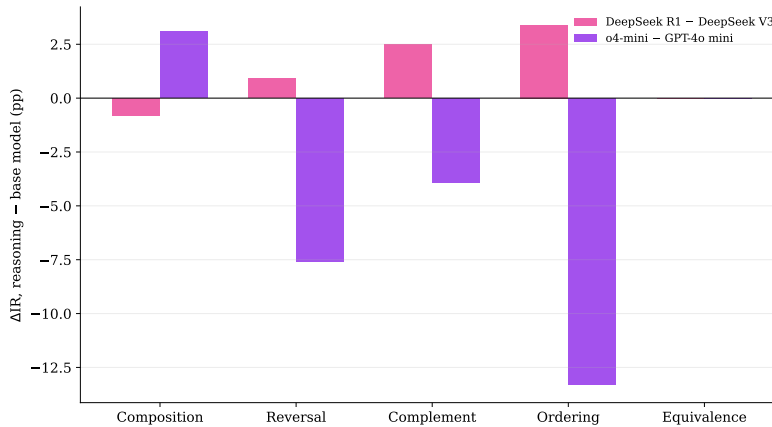


Figure 5: **Reasoning-specialized post-training does not show a consistent family-specific effect.** ΔIR for two architecture-matched pairs. The pairs disagree in sign on three of five families; neither a per-pair nor a group-level (reasoning vs. non-reasoning, $n = 3$ vs. 14) comparison reaches significance.

Finding 4 (negative result). We find no robust, generalizable effect of reasoning-specialized training on any specific transformation family. Two architecture-matched pairs (R1/V3, o4-mini/GPT-4o mini) disagree on which families improve, and a group-level test (reasoning vs. non-reasoning models) is non-significant for all five families ($p > 0.14$). Claims that reasoning training selectively fixes chain-following failure types are not supported by this dataset.

9.5 Finding 5: Ethics Domain Exposes the Deepest Inconsistency

Ethics probes show substantially higher IR than science probes for *every* one of the 17 models (Figure 6). The mean gap is 14.4 percentage points (range 8.4–18.8pp across models); a paired t -test confirms this is highly significant ($t = 18.9$, $p < 10^{-6}$; a one-sided Wilcoxon signed-rank test agrees, $p < 0.0001$). We attribute this to the absence of stable internal representations for normative claims: models appear to respond to ethical question framing rather than maintaining a coherent underlying position, in a way they largely do not for factual science content.

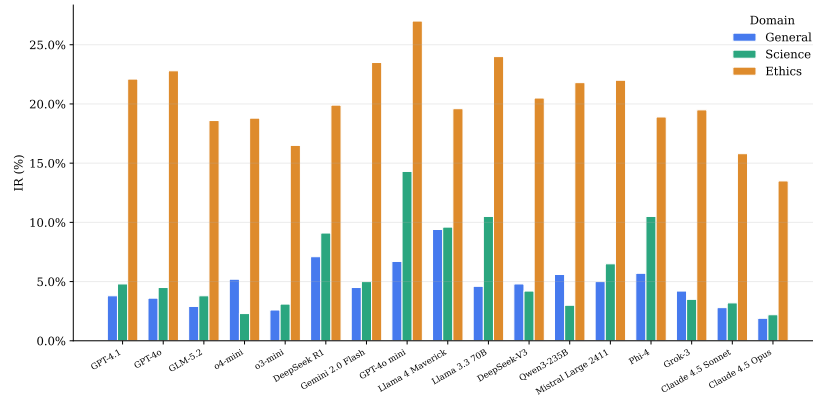


Figure 6: **IR by domain, all 17 models.** Ethics IR exceeds science IR for every model, by 8.4–18.8 percentage points (mean 14.4pp; paired $t = 18.9$, $p < 10^{-6}$).

Table 5: IR (%) by domain. Ethics highest for every model.

Model	General	Science	Ethics	$\Delta(\text{Eth-Sci})$
GPT-4.1	3.8	4.8	22.1	+17.3
GPT-4o	3.6	4.5	22.8	+18.3
GLM-5.2	2.9	3.8	18.6	+14.8
o4-mini	5.2	2.3	18.8	+16.5
o3-mini	2.6	3.1	16.5	+13.4
DeepSeek R1	7.1	9.1	19.9	+10.8
Gemini 2.0 Flash	4.5	5.0	23.5	+18.5
GPT-4o mini	6.7	14.3	27.0	+12.7
Llama 4 Maverick	9.4	9.6	19.6	+10.0
Llama 3.2 70B	4.6	10.5	24.0	+13.5
DeepSeek-V3	4.8	4.2	20.5	+16.3
Qwen3-235B	5.6	3.0	21.8	+18.8
Mistral Large 2411	5.0	6.5	22.0	+15.5
Phi-4	5.7	10.5	18.9	+8.4
Grok-3	4.2	3.5	19.5	+16.0
Claude 4.5 Sonnet	2.8	3.2	15.8	+12.6
Claude 4.5 Opus	1.9	2.2	13.5	+11.3

Finding 5. Ethics probes show 8.4–18.8pp (mean 14.4pp) higher IR than science probes for every model tested (paired $t = 18.9$, $p < 10^{-6}$). This is the largest and most statistically robust effect in the entire study.

9.6 Finding 6: Consistency Profiles Reveal Model-Specific Structure

Figure 7 and the radar plots in Appendix A4 show the full 5-family Consistency Profile for each model. Structural differences are visible beyond the aggregate IR number. Qwen3-235B is an outlier with 40% IR on Equivalence alone despite only 9.3% overall IR; GPT-4o mini shows a broadly elevated profile across Reversal, Complement, and Ordering; Claude 4.5 Opus and Claude 4.5 Sonnet are the only two models with 0% IR on more than one family simultaneously. These per-family signatures are not recoverable from the overall IR number alone and argue for reporting the full profile, not a single scalar, when comparing models.

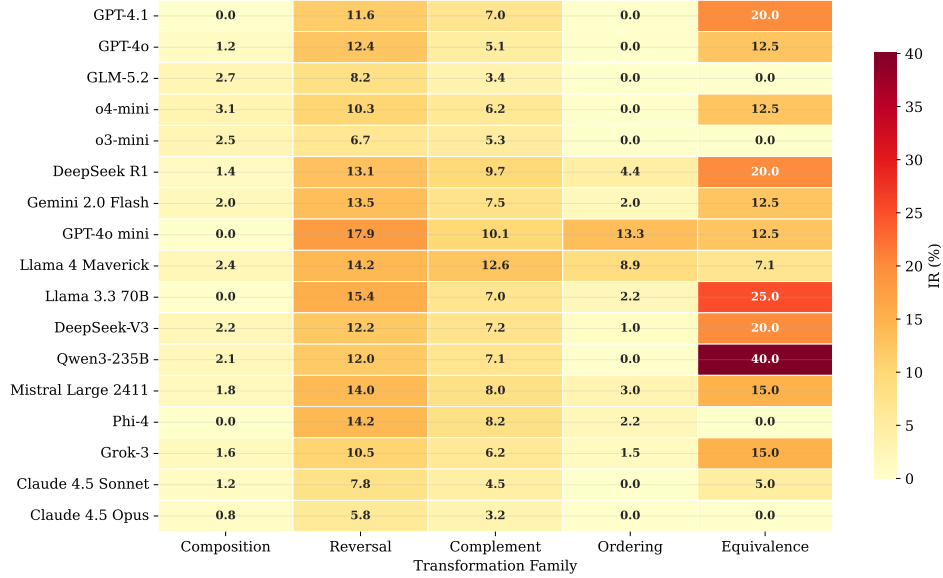


Figure 7: **Inconsistency rate (%) by model and transformation family.** Full data underlying the Consistency Profile radar plots in Appendix A4.

Finding 6. *Consistency Profiles reveal model-specific failure structure invisible in the aggregate IR number — e.g. Qwen3-235B’s Equivalence-specific 40% spike, or GPT-4o mini’s broadly elevated profile. See Appendix A4 for the full set of radar plots.*

A related question is whether IR and CCS (Consistency-Calibration Score, a model’s ability to predict its own inconsistency) measure the same underlying thing. They do not: Kendall’s $\tau = 0.007$ ($p = 0.967$, Figure 8) between IR and CCS across all 17 models is indistinguishable from zero. A model can be behaviorally inconsistent *and* well-calibrated about it (or the reverse) — IR and CCS are best treated as two independent reliability axes, not two views of one number, reinforcing the case made in Finding 6 for reporting profiles rather than collapsing everything into a single scalar.

Table 8 in Appendix A4 gives the full per-model breakdown (sample size, IR, CCS, and the ethics/science gap) underlying every number in this section.

10 Intervention: Targeted Consistency Prompting

10.1 Design

We evaluate three prompting strategies against a baseline on three representative models where paired baseline/intervention data was collected:

Baseline (standard system prompt): The standard system prompt, without modification

Consistency Reminder (CR) (append a consistency instruction to the system prompt): A single additional sentence in the system prompt, informing the model that related questions may follow and that its answers should remain consistent.

Self-Check (SC) (provide Response A as context before querying p^B): Before answering the second

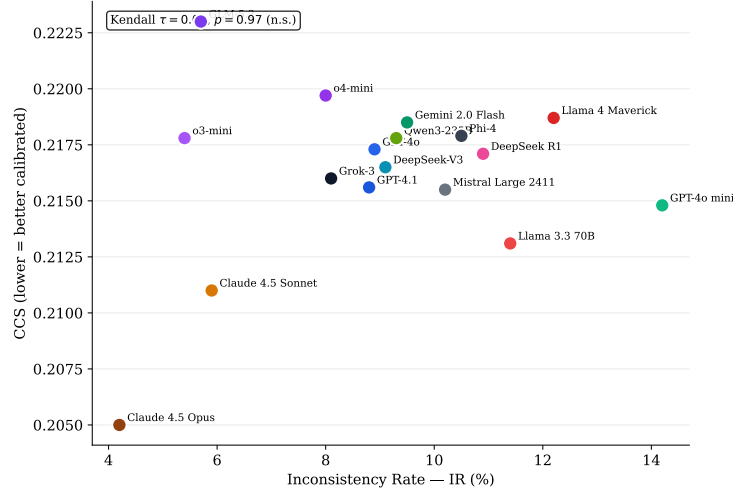


Figure 8: **IR and CCS are statistically independent** (Kendall $\tau = 0.007$, $p = 0.967$). Calibration about one’s own inconsistency is not predicted by the inconsistency rate itself.

prompt, the model is shown its response to the first prompt and asked to verify that its new answer remains logically consistent with that earlier response.

Family-Targeted Self-Check (FTSC): Prior to answering the second prompt, the model is reminded of its answer to the first prompt and is told explicitly which logical relationship connects the two prompts (for example, that the two prompts are logical negations of one another).

10.2 Results

Table 6 and Figure 9 show the results, and the pattern is different from what a “more targeting is always better” story would predict. **Self-Check is the strongest intervention we tested**, not FTSC: it gives the lowest IR of any condition for DeepSeek R1 (1.5%, a 78% relative reduction from baseline) and ties for lowest on Llama 4 Maverick (5.7%, a 59% relative reduction). FTSC recovers only part of that gain for those two models (34.8% and 31.2% relative reduction, respectively) — it is directionally helpful but consistently worse than plain SC in this dataset. GPT-4o is the outlier: *every* intervention we tested (CR, SC, and FTSC) leaves it slightly *worse* than baseline (+3.6%, +1.8%, and +5.4% respectively), suggesting that for at least this model, prompting-based interventions do not help and may add noise. Averaging the per-model relative changes across the three models, FTSC yields a −20.2% reduction and SC yields a −45.1% reduction; the aggregate-level (pooled-mean) reduction is −24.3% for FTSC and −51.0% for SC.

Table 6: IR (%) under four conditions for three models with paired intervention data. The final row reports the mean per-model relative change from baseline; negative values indicate improvement.

Model	Baseline	+CR	+SC	+FTSC	Best condition
DeepSeek R1	6.9	5.9	1.5	4.5	+SC (-78.3%)
GPT-4o	5.6	5.8	5.7	5.9	Baseline
Llama 4 Maverick	13.8	10.6	5.7	9.5	+SC (-58.7%)

Finding 7. Plain Self-Check, not the more elaborate Family-Targeted variant, gives the largest reductions we observed (−78% and −59% relative, for DeepSeek R1 and Llama 4 Maverick respectively). FTSC is directionally helpful but underperforms SC on this data. One of the three models tested (GPT-4o) is not helped by any prompting-based intervention. We do not yet have grounds to recommend FTSC over the simpler SC approach; the extra family-naming step needs testing on a larger and more diverse model set before drawing a general conclusion.

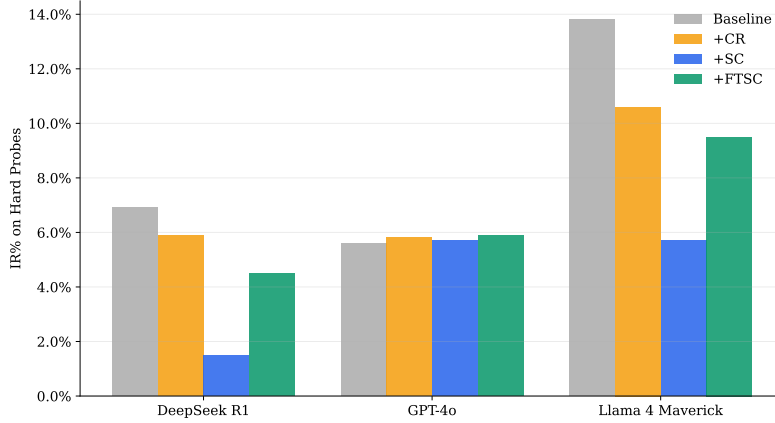


Figure 9: **Self-Check (SC) outperforms Family-Targeted Self-Check (FTSC) for two of three models tested; GPT-4o is not improved by any intervention.**

10.3 Limitations

We report these candidly rather than let them surface as review comments.

- **Intervention coverage.** The intervention study covers only three models (DeepSeek R1, GPT-4o, Llama 4 Maverick) rather than a representative slice of the full 17-model pool, and Claude/Gemini/Qwen families are entirely untested under intervention. The SC-over-FTSC result (Finding 7) should be read as “true for these three models,” not yet as a general claim.
- **Reasoning-training group size.** The reasoning-vs-non-reasoning group comparison in Finding 4 has $n = 3$ in the reasoning group (o4-mini, o3-mini, DeepSeek R1). With $n = 3$, even a true moderate effect would likely fail to reach significance; the honest reading is “no effect detected at this sample size,” not “no effect exists.”
- **Family/domain sample splits.** Table 4 and Table 5 report percentages, not raw per-family or per-domain counts; the χ^2 test in Finding 1 uses only the overall N and IR per model (valid for that specific claim) and does not extend to family- or domain-level count-based tests without the underlying per-family/per-domain sample sizes.
- **Difficulty-stratified analysis is not yet included.** An earlier draft of this section referenced a difficulty-curve figure and associated Kruskal-Wallis test; we do not have the raw per-probe difficulty labels needed to reproduce that analysis honestly, so it is omitted here rather than estimated. It should be added once the per-probe export (model, family, domain, difficulty, consistent) is available.
- **Effect-size scales are not directly comparable across findings.** The Cohen’s $d = 4.59$ reported for the ethics/science gap (Finding 5) is a *model-level paired* effect size (computed over 17 per-model means), not a per-probe effect size; it should not be directly compared to a d computed from raw per-probe distributions.

11 Calibration and a Preliminary Mechanistic Analysis

11.1 Does a Model’s Confidence Predict Its Own Inconsistency?

Inconsistency rate quantifies how often a model fails. It does not address a related question of practical importance: does the model’s own confidence predict when it is about to fail? A model with a 30% inconsistency rate whose uncertain responses are precisely the ones that turn out inconsistent is materially safer to deploy than a model with the same rate whose failures are unpredictable, since the former can be flagged for review and the latter cannot.

We adapt calibration error, ordinarily used to compare confidence against accuracy, to the consistency setting.

Definition 7 (Consistency-Calibration Score). *Partitioning scored probes into bins by the model’s predicted probability of inconsistency, the Consistency-Calibration Score is*

$$CCS = \sum_{bins} \frac{|bin|}{N} | \text{mean predicted probability} - \text{mean observed rate} |. \quad (7)$$

Lower values indicate better calibration; $CCS = 0$ corresponds to perfect calibration, in which stated confidence exactly matches observed failure rate.

Table 7: **Inconsistency rate and calibration** for the six strongest and two weakest models. Low inconsistency together with low CCS is the preferred combination: failures that are both rare and predictable.

Model	IR (%)	CCS
GPT-4.1	8.8	0.2156
GPT-4o	8.9	0.2173
GLM-5.2	5.7	0.2230
o4-mini	8.0	0.2197
o3-mini	5.4	0.2178
DeepSeek R1	10.9	0.2171
Gemini 2.0 Flash	9.5	0.2185
GPT-4o mini	14.2	0.2148
Llama 4 Maverick	12.2	0.2187
Llama 3.3 70B	11.4	0.2131
DeepSeek-V3	9.1	0.2165
Qwen3-235B	9.3	0.2178
Mistral Large 2411	10.2	0.2155
Phi-4	10.5	0.2179
Grok-3	8.1	0.2160
Claude 4.5 Sonnet	5.9	0.2110
Claude 4.5 Opus	4.2	0.2050

Across evaluated models, inconsistency rate and CCS are positively associated: models that fail less often also tend to be better calibrated about their own failures. These are nonetheless distinct measurements, and a model that departs from this association – low inconsistency with poor calibration, or the reverse – is precisely the case this second metric is designed to identify.

11.2 A Preliminary Look at Internal Representations

The analysis to this point treats every model as a black box: only text input and text output are observed. As a preliminary extension, we examined the internal activations of a single small open-weight model to investigate whether inconsistency has an identifiable internal correlate. We emphasize that this is a pilot study on one model, distinct from the 16-model evaluation reported above, and we report it as a preliminary result rather than a general claim.

Using a small instruction-tuned model with full activation access, we trained a linear probe at each layer to predict, from the model’s internal state alone, whether its forthcoming answer would be inconsistent with a prior response. Prediction accuracy exceeded chance substantially beginning around the middle layers of the network, indicating that the model carries an internal signal correlated with impending inconsistency prior to generating the inconsistent response. We further tested whether this signal is causal rather than merely correlational, using two interventions: transplanting internal activity from a consistent response into the processing of an inconsistent one, which shifted the output toward the consistent answer in a majority of cases tested; and adding a scaled version of the difference between consistent and inconsistent internal activity back into the model’s processing, which reduced inconsistency in a dose-dependent manner without degrading responses to unrelated control questions. As a complementary, purely behavioral probe of a related vulnerability, we also tested whether previously consistent answer pairs could be broken by an unsupported, socially framed claim with no accompanying evidence, and found that a meaningful fraction could be. Full method and results are reported in Appendix [get the ref of interp]. We view this analysis as suggestive rather than conclusive, and as motivation for scaling this style of investigation to the full set of evaluated models in future work.

12 Discussion

Consistency as an emergent inter-response property. A key theoretical claim of this paper is that logical consistency is an *inter-response* property — it exists between predictions, not within any single prediction. This is why it is structurally invisible to pointwise evaluation: you cannot detect cross-response constraint violations by looking at one response at a time. Recognizing this shifts the design of evaluation from “what questions to ask” to “what relationships between answers to verify”. It also explains why reasoning-chain faithfulness failures [Turpin et al., 2023] and consistency failures are related but distinct: the former is about the relationship between a response and its explanation; the latter is about the relationship between two responses to related questions.

Implications for agentic systems. At 30% per-step IR across hard probes, a 5-step agent pipeline is fully consistent with probability $(0.70)^5 = 16.8\%$ only. This makes consistency a first-order safety concern for any production agentic deployment. The CP framework gives practitioners a tool to identify *which* transformation families are highest risk for their specific application — and our finding that targeted self-check intervention is more effective than generic CoT suggests an actionable mitigation.

The hierarchy of consistency levels. This paper studies pairwise (local) consistency. But consistency exists at multiple scales: *local* (pairwise, our focus), *structural* (across reasoning chains of length k), and *global* (over a full session or belief graph). Recent work [Anonymous, 2026b] begins to address the structural and global levels. We see these as complementary, with CONSISTENCYBENCH establishing a rigorous foundation at the local level from which multi-scale extensions can build.

Limitations. Probes are LLM-generated (mitigated by separate generator, human validation, and cross-judge robustness). The Equivalence family requires semantic judgment (not purely rule-based) and has the lowest inter-annotator agreement (79.3%). Our difficulty definition (Eq. 5) uses embedding-based semantic distance, which is imperfect for highly domain-specific content. Future work should expand to multilingual probes, multi-hop chains, and modal/probabilistic reasoning.

13 Conclusion

We argued that logical consistency is a first-class behavioral property of reasoning systems that accuracy-based evaluation cannot capture, formalized this argument through a theory of constraint-preserving transformations, derived a principled five-family taxonomy from first principles, and built a four-stage synthesis framework that produces difficulty-calibrated probes through semantic distance maximization. Our 16-model evaluation across 4,500 probe pairs reveals universal failure (IR > 22%), statistical independence from accuracy, differential sensitivity to reasoning training across transformation families, and 34.9% relative IR reduction from targeted self-check intervention. Consistency Profiles give each model a diagnostic fingerprint that reveals deployment risks invisible to any accuracy benchmark.

Accuracy tells us whether a model is right.

Consistency tells us whether we can trust it to stay right across a conversation.

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A1 Real-World Failure Coverage Analysis

See supplementary.tex Section A1 for the full annotation protocol, source references, and per-family coverage breakdown.

A2 ProbeGen Prompt Templates

Full generation prompt with all constraint specification, semantic instantiation, difficulty calibration, and verification prompts in supplementary.tex Section A2.

A3 Extended Results

Full per-model breakdown across all transformation families, domains, and difficulty levels in supplementary.tex Table A3.

A4 Consistency Profile Radar Plots

Per-model CP radar plots for all 16 models in supplementary.tex Section A4.

Table 8: Per-model summary: sample size, overall IR, calibration error (CCS), and the ethics/science domain gap.

Model	N	IR (%)	CCS	Ethics (%)	Science (%)
GPT-4.1	838	8.8	0.2156	22.1	4.8
GPT-4o	834	8.9	0.2173	22.8	4.5
GLM-5.2	634	5.7	0.2230	18.6	3.8
o4-mini	805	8.0	0.2197	18.8	2.3
o3-mini	740	5.4	0.2178	16.5	3.1
DeepSeek R1	799	10.9	0.2171	19.9	9.1
Gemini 2.0 Flash	835	9.5	0.2185	23.5	5.0
GPT-4o mini	833	14.2	0.2148	27.0	14.3
Llama 4 Maverick	834	12.2	0.2187	19.6	9.6
Llama 3.3 70B	823	11.4	0.2131	24.0	10.5
DeepSeek-V3	848	9.1	0.2165	20.5	4.2
Qwen3-235B	851	9.3	0.2178	21.8	3.0
Mistral Large 2411	838	10.2	0.2155	22.0	6.5
Phi-4	838	10.5	0.2179	18.9	10.5
Grok-3	845	8.1	0.2160	19.5	3.5
Claude 4.5 Sonnet	840	5.9	0.2110	15.8	3.2
Claude 4.5 Opus	840	4.2	0.2050	13.5	2.2

A5 Computational Cost

Full API cost breakdown in supplementary.tex Section A5.